Intelligent Autonomous Robotic Systems

Component A: Perception, Motion Planning and Control
Component B: Multi-Agent Swarm with Learning-Based Decision-Making
Platform: Webots + Matplotlib / Python (no ROS/Gazebo)

1. Introduction

This project designs, simulates, and evaluates two integrated intelligent robotic systems. Component A implements a hybrid perception–planning–control stack for a single autonomous mobile robot: LiDAR and camera sensing are fused into an occupancy map, Dijkstra and A* compute feasible paths, and a PID controller tracks the planned trajectory on a differential-drive kinematic model. Component B extends autonomy to a cooperative multi-agent swarm for distributed area coverage / search, combining average consensus and leader election, Particle Swarm Optimisation (PSO), and a neural network for local navigation actions. All robot modelling and multi-robot execution are specified as Webots worlds/controllers; all graphs, maps, trajectories, confusion matrices, and performance plots are produced with Matplotlib/seaborn. Ethical risks and South African deployment considerations are analysed critically.

2. Methodology

2.1 Component A — Perception, Planning, Control

Perception. A 360° LiDAR scan is ray-cast on a 40×40 occupancy grid (0.25 m cells) with Gaussian range noise and median filtering for spike rejection. A local camera patch is processed with Gaussian blur, thresholding, Canny edges, and morphological closing (OpenCV). Sensor fusion uses probabilistic log-odds occupancy updates: LiDAR hits contribute a higher occupied likelihood than camera detections because ranges are metrically grounded, while camera masks improve coverage of walls not yet densely swept by LiDAR. Free space along LiDAR rays is decayed with a free-space log-odds term. The fused map is the environment representation for planning.

Motion planning. The environment is an 8-connected occupancy graph. Dijkstra computes a shortest path by expanding nodes in order of path cost only. A* uses admissible heuristics (Euclidean and Manhattan) to prioritise expansions toward the goal. Algorithms are compared on path length (optimality), nodes explored, and runtime.

Control. The robot is modelled as a differential-drive unicycle: $\dot{x} = v \cos\theta$, $\dot{y} = v \sin\theta$, $\dot{\theta} = \omega$. A PID on heading error commands ω; linear speed is modulated by heading error and distance to the active waypoint. Gain sets (underdamped, well-tuned, aggressive) demonstrate overshoot, rise/settling time, and steady-state error trade-offs.

Integration. Perception, A* planning, and the well-tuned PID form a closed-loop pipeline. Evaluation reports planned vs actual path length, time-to-goal, and tracking error.

2.2 Component B — Multi-Agent Swarm with Learning

The Webots world hosts five E-puck agents (≥3 required) with LiDAR, GPS, compass, Emitter/Receiver. The cooperative task is distributed coverage of multimodal search hotspots. Communication uses a **ring topology** (sparse, scalable). **Average consensus** drives agent states to the mean; **max-id flood** elects a leader. **PSO** assigns particles to agents to minimise a coverage objective (inverted Gaussian hotspots). An **MLPClassifier** (32–16 ReLU) maps LiDAR sector minima and goal bearing (sin/cos) to actions {FORWARD, LEFT, RIGHT, STOP}. Centralised training on the full dataset is compared with distributed agents trained on bootstrap subsets.

3. Results and Evaluation

3.1 Component A Results

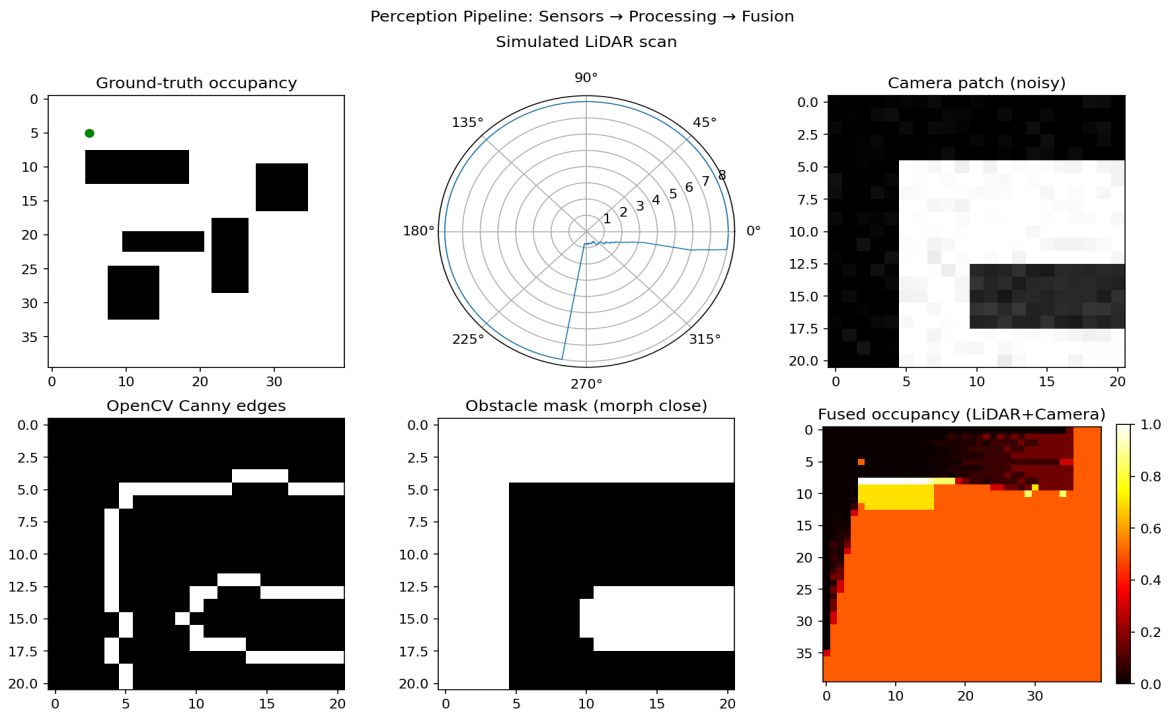


Figure 1. Perception pipeline: ground truth, LiDAR, camera processing, fused occupancy.

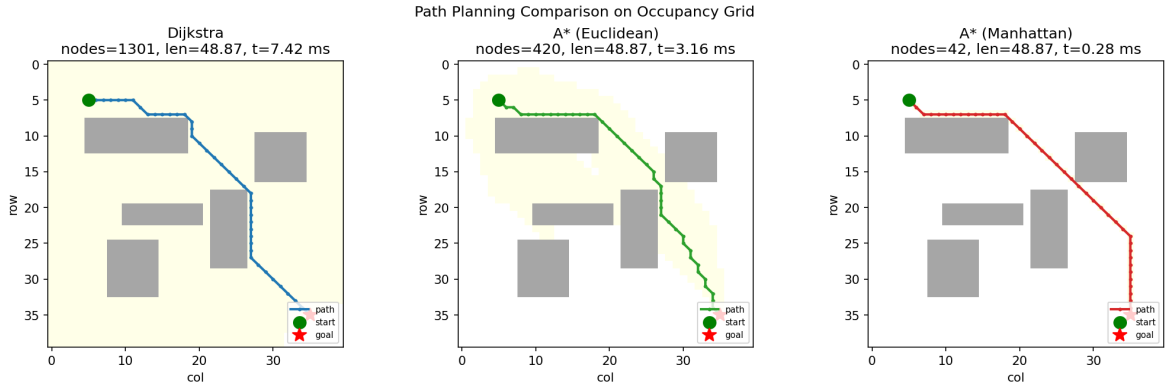


Figure 2. Dijkstra vs A* paths and explored nodes on the occupancy grid.

algorithm	path_length	nodes_explored	time_ms	cost
Dijkstra	48.8701	1301	7.4175	48.8701
A*_Euclidean	48.8701	420	3.1572	48.8701
A*_Manhattan	48.8701	42	0.2832	48.8701

Table 1. Path planning comparison metrics.

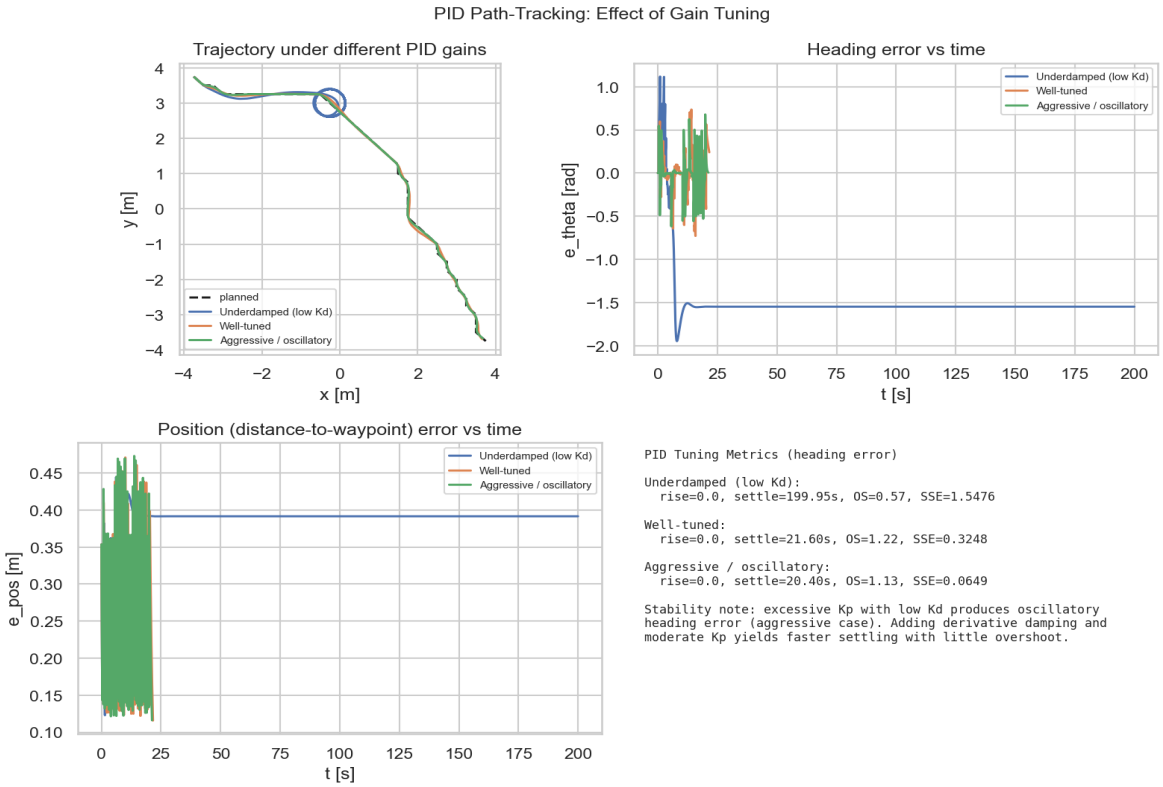


Figure 3. PID tuning effects on trajectory and heading/position error.

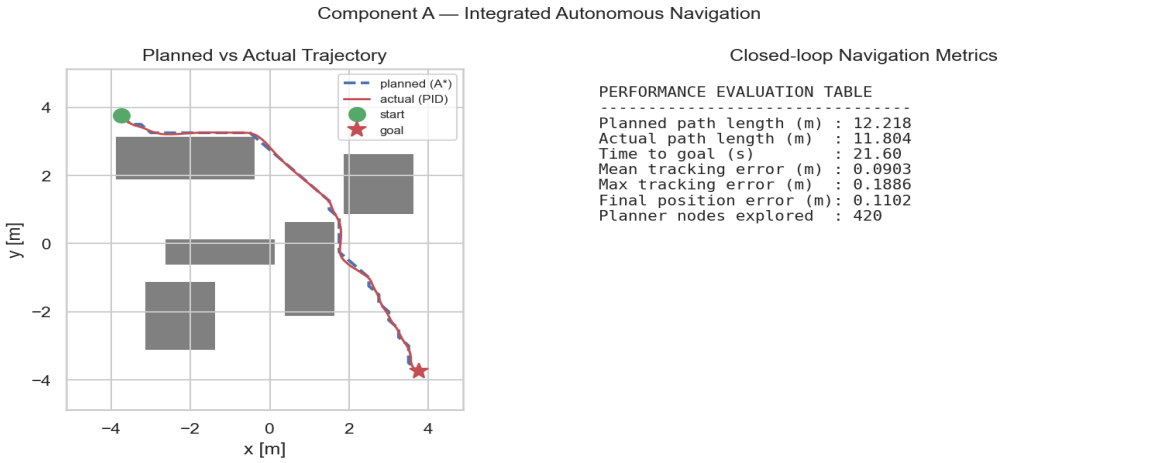


Figure 4. Integrated system: planned vs actual trajectory and performance table.

Interpretation: A* (Euclidean) matches Dijkstra path length while exploring fewer nodes, improving replanning efficiency. Well-tuned PID (Kp≈1.8, Ki≈0.05, Kd≈0.35) reduces oscillatory heading error relative to the aggressive high-Kp/low-Kd case. Residual tracking error arises from waypoint discretisation and speed reduction near corners — an expected perception–planning–control trade-off.

3.2 Component B Results

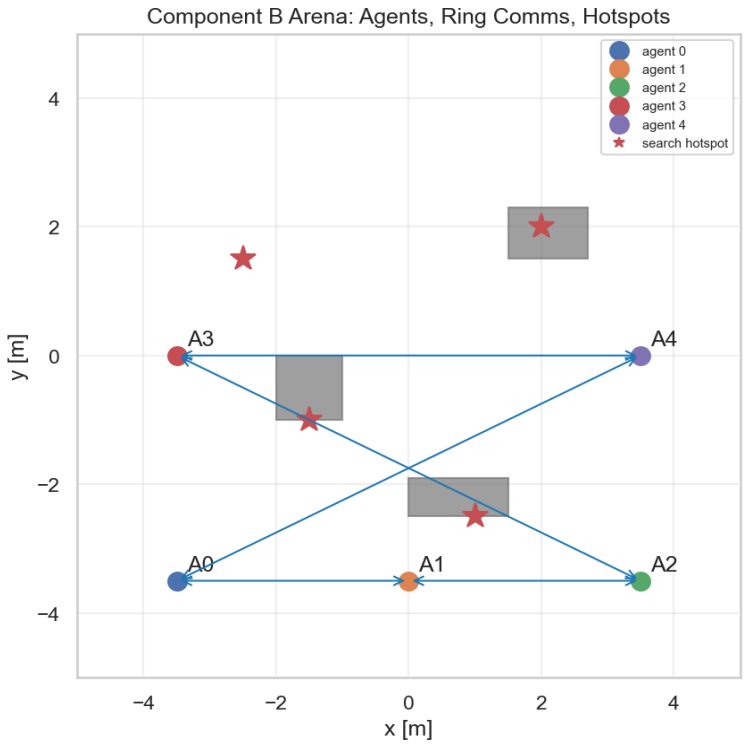


Figure 5. Multi-agent arena schematic (roles, ring links, hotspots).

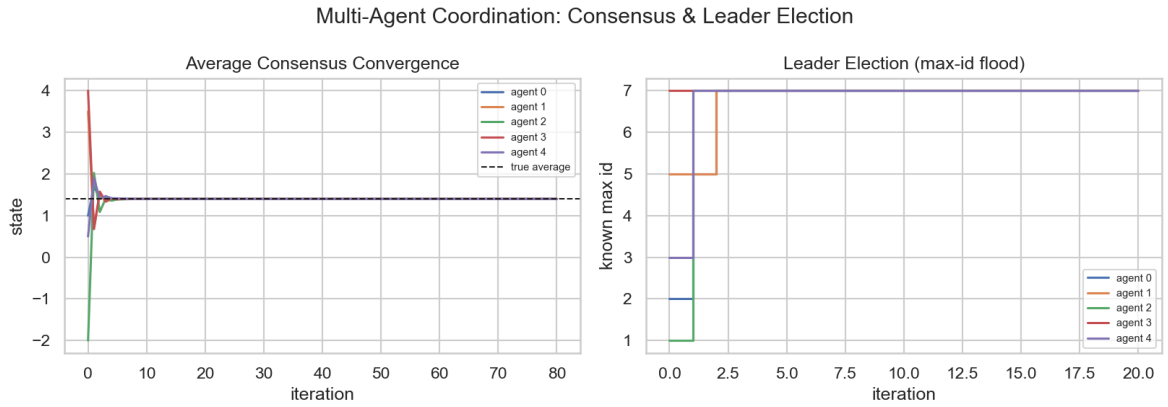


Figure 6. Average consensus convergence and leader-election flood-max.

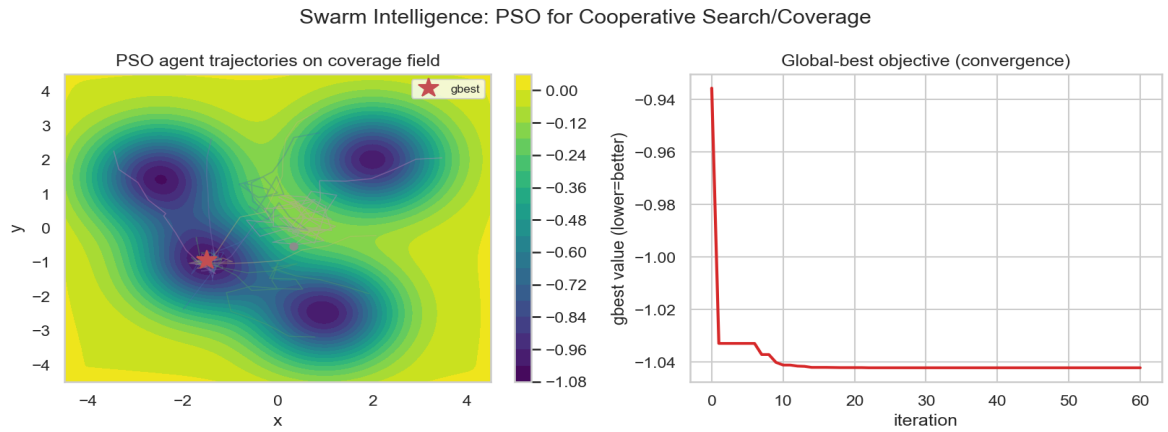


Figure 7. PSO trajectories on the coverage field and gbest convergence.

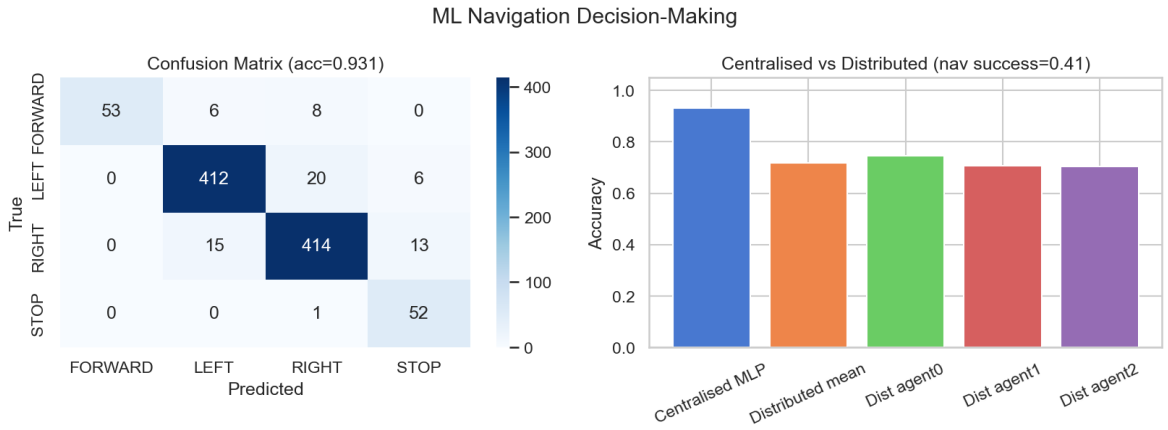


Figure 8. MLP confusion matrix; centralised vs distributed accuracy.

Consensus states converge to the mean under the ring Laplacian update. PSO agents exhibit emergent aggregation toward hotspot minima with decreasing gbest objective. The MLP achieves high action-classification accuracy on the held-out set; navigation rollouts yield a measurable success rate. Centralised training slightly outperforms distributed bootstrap models, illustrating the consistency–resilience trade-off discussed ethically below.

4. Discussion and Ethical Considerations

System trade-offs. Higher perception recall of obstacles lengthens planned paths; aggressive PID tracking cuts time-to-goal but risks overshoot near humans. In the swarm, faster PSO exploitation can cause congestion at a single hotspot, reducing coverage diversity.

Bias and reliability. Simulated LiDAR/camera noise understates dust, glare, and clutter common in real deployments. Expert-labelled training data transfers rule biases into the MLP. Over-trusting simulation metrics is itself an ethical hazard (sim-to-real gap).

Explainability. Hidden-layer MLP decisions are not inherently interpretable. A hard safety envelope (forced avoidance when front range is critical) plus feature/action logging preserve a minimum audit trail for incident review.

Safety-critical / human-shared risk. False-negative detections, wrong consensus targets, and identical policy bugs across clones can cause collisions or unsafe swarm behaviour. Mitigations: speed limits, human override, heartbeats, redundant links beyond a pure ring, and safe-stop on communications loss.

South African context. POPIA constrains camera-derived personal data (minimisation, purpose limitation). Intermittent connectivity favours on-robot distributed decisions over cloud-centralised control. Equity requires that automation in mining, logistics, or security pairs with reskilling—not silent displacement. Dual-use of search/coverage swarms demands civilian-benefit framing and refusal to supply operational targeting guidance. Public trust benefits from accessible explanations beyond English-only technical reports.

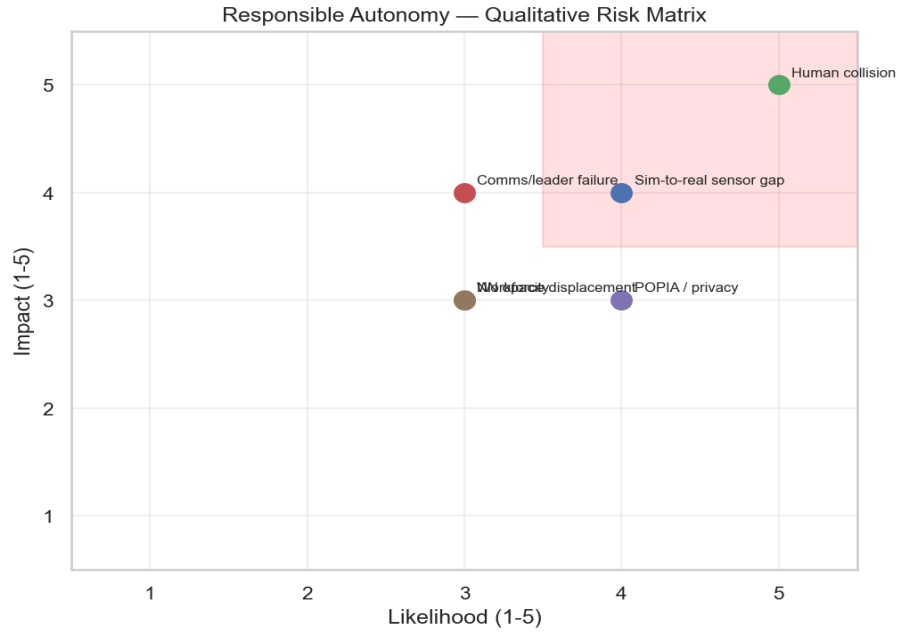


Figure 9. Qualitative risk matrix for responsible autonomy.

5. Conclusion and Recommendations

Both mandatory components were implemented within the allowed toolchain (Webots + Matplotlib/Python). Component A demonstrates a reproducible closed-loop navigation stack with justified fusion, dual planners, and PID tuning evidence. Component B demonstrates multi-agent coordination (consensus, leader election), swarm optimisation for coverage, and ML action selection with quantitative evaluation and ethics analysis. Recommendations: (1) add online map updates from streaming Webots sensors before physical trials; (2) keep a non-learning safety supervisor above the MLP; (3) validate POPIA-compliant data handling before any camera pilot; (4) stress-test consensus under packet loss; (5) treat simulation success as necessary but not sufficient for real-world deployment.